

Agam Katiyar

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EDUCATION

B.Tech in Computer Science Engineering (AI & ML)

Vellore Institute of Technology, Bhopal

Aug 2023 – Aug 2027

CGPA: 8.56/10

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript
- **Frontend:** React.js, HTML5, CSS3
- **Backend:** Node.js, REST APIs
- **Databases:** MySQL, MongoDB
- **AI/ML:** TensorFlow, OpenCV, Whisper, MediaPipe, LangChain, LangGraph
- **Core CS:** Data Structures & Algorithms, OOP
- **Tools:** Git, GitHub

PROJECTS

Real-Time Speech-to-Speech Translator | Python, Whisper, NLLB-200, Tkinter

[GitHub](#)

- Built a fully offline multilingual speech-to-speech translation system using Whisper STT and Meta NLLB-200.
- Integrated real-time audio streaming, translation, and TTS pipelines with multithreaded processing for responsive UI performance.
- Optimized CPU inference and model execution using PyTorch, OpenMP tuning, and low-latency audio processing.

H-GestureControl: Hand Gesture Laptop Controller | Python, OpenCV, MediaPipe

[GitHub](#)

- Developed a real-time hand gesture recognition system for touch-free laptop control using computer vision.
- Enabled cursor movement, volume control, and window switching through webcam-based gesture tracking.
- Integrated MediaPipe hand landmarks and PyAutoGUI for low-latency system interaction and improved accessibility.

Driver Drowsiness Detection System | Python, OpenCV, MediaPipe

[GitHub](#)

- Built a real-time driver monitoring system using EAR/MAR-based facial landmark analysis for drowsiness detection.
- Implemented MediaPipe Face Mesh and real-time alert logic for eye-closure and yawn detection on CPU-only systems.
- Developed audio-visual alert mechanisms and live monitoring dashboards for enhanced driving safety.

EXTRA-CURRICULAR AND ACHIEVEMENTS

Hackathon and Coding

- Participated in multiple AI/ML-focused hackathons centered on real-world problem solving.
- **GRABHACK 2025 Pre-Finalist:** Ranked among Top 98 teams out of 4500+ teams nationwide for building an AI-driven disruption management system.
- **AI/GenAI Hackathon 2025:** Proposed an AI-powered deepfake video detection system using machine learning and deep learning techniques.
- **National CyberShield Hackathon:** Proposed an AI-based fraud detection model for real-time suspicious transaction flagging using ML and feature engineering.
- Solved 300+ DSA problems across LeetCode, Codeforces, and GeeksforGeeks.

Volunteer and Social Work

- **Event Volunteer, NSS VIT Bhopal** – Contributed to the “10,000 Mega Tree Plantation in a Day” initiative (Sehore District Forest Area, Aug 2024), supporting large-scale environmental sustainability efforts.
- Earned Google Arcade swags through successful completion of Google Cloud labs and quizzes.

CERTIFICATIONS

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** – Oracle
- **Applied Machine Learning in Python** – University of Michigan (Coursera)
- **Google AI Essentials** – Google
- **Data Engineering Foundation** – Informatica